Osmotic Shock Transformation of D. salina

To get our target genes expressed in Dunaliella salina, we used an osmotic shock transformation method to deliver the assembled plasmids into the cells.

This section includes:

1. Cell culture and preconditioning
2. Screening and identification of transformants

Protocol – Osmotic Shock Transformation of D. salina

Cell culture and preconditioning

D. salina FACHB-435 was grown statically in modified DM medium (with 1.5 M NaCl) at 26°C, under 50 μmol photons m^{-2} s^{-1} of light and a 16 h daily photoperiod. The cells were harvested once they reached logarithmic phase (about 10^5–10^6 cells/mL), at which point the OD680 reads roughly 1.0.

1. Take the log-phase culture and pellet the cells in a centrifuge.
2. Pour off the supernatant, resuspend the pellet in fresh PKS/DM medium, and wash the cells 3 times to strip off residual metabolites from the old medium.
3. Resuspend the washed cells in fresh medium and adjust the final cell density to 10^6 cells/mL.
4. Transfer this suspension into fresh medium containing 1 M NaCl and keep culturing for another 8 h. This step "pre-adapts" the cells to the osmotic shock that follows.
5. Take 1 mL of the preconditioned suspension, spin it down, and discard the supernatant.
6. Immediately resuspend the pellet in fresh DM medium containing 0.1 M NaCl (bringing the volume back to 1 mL).
7. Add the following to this 1 mL suspension, in order and precisely:
    a) Plasmid DNA: 5.46 μg of the vector carrying the target gene
    b) Surfactant: 15 μL of 0.1% Triton X-100 stock
8. Vortex the mixture vigorously for about 90 s.
9. As soon as the vortexing is done, transfer the mixture into fresh liquid medium.
10. Let the cells recover statically for 48 h at 26°C under a 16 h daily photoperiod. By then, over 90% of the surviving transformed cells can return to normal growth.

Screening and identification of transformants

Fluorescence microscopy: cells from the recovery culture were collected and fixed with 4% paraformaldehyde for 30 min, washed twice by resuspension in PBS, and then examined under a fluorescence microscope. Because the vector carries EGFP, successfully transformed cells show green fluorescence.

Molecular-level identification (PCR): genomic DNA was extracted from the transformed cells and amplified with gene-specific primers. A band of the expected size confirms that the exogenous gene has been stably integrated into the cell genome.

Selection steps:

1. DM solid medium with a salt concentration of 0.5 M and 2% agar
2. Antibiotic concentration screen: take the same amount of D. salina as used after the osmotic transformation culture, collect the algal pellet by low-speed centrifugation, and plate it on D. salina plates with different antibiotics to pin down the minimum lethal concentration
3. Plate the post-transformation algae onto plates with the corresponding antibiotic; after they have grown for a while, pick single colonies and scale them up — these are the transformants

4. Extract the genomic DNA again and verify the transformants by qPCR

Plant Genomic DNA Extraction

To check whether the target gene had actually integrated into the chromosomal genome of D. salina, we extracted genomic DNA from the algae.

This section includes:

1. DNA extraction
2. DNA quantification and purity check

Protocol –Plant Genomic DNA Extraction

Genomic DNA extraction

We used the Plant Genomic DNA Kit from TIANGEN.

Before starting, add anhydrous ethanol to buffer GD and wash buffer PW — the volumes are specified on the bottle labels.

1. Take about 100 mg of fresh D. salina algal pellet (or ~30 mg of dried tissue) and grind it thoroughly in liquid nitrogen.
2. Quickly transfer the ground powder into a centrifuge tube pre-loaded with 700 μL of buffer GP1 pre-warmed to 65°C (β-mercaptoethanol was added to the pre-warmed GP1 beforehand, to a final concentration of 0.1%). Mix right away by inverting, then incubate the tube in a 65°C water bath for 20 min, inverting the tube a few times along the way to keep the sample mixed.
3. Add 700 μL of chloroform, mix thoroughly, and centrifuge at 12,000 rpm (~13,400 × g) for 5 min. Note: for tissues rich in polyphenols or starch, do an equal-volume phenol:chloroform (1:1) extraction before this step.
4. Carefully transfer the upper aqueous phase from the previous step into a fresh centrifuge tube, add 700 μL of buffer GP2, and mix well.
5. Load the mixture onto adsorption column CB3, centrifuge at 12,000 rpm (~13,400 × g) for 30 s, and discard the flow-through. (The column holds about 700 μL, so it can be loaded and spun in batches.)
6. Add 500 μL of buffer GD to column CB3 (double-check that anhydrous ethanol has been added), centrifuge at 12,000 rpm (~13,400 × g) for 30 s, pour off the waste, and put the column back into the collection tube.
7. Add 600 μL of wash buffer PW to column CB3 (double-check that anhydrous ethanol has been added), centrifuge at 12,000 rpm (~13,400 × g) for 30 s, pour off the waste, and put the column back into the collection tube.
8. Repeat step 7.
9. Put column CB3 back into the collection tube and centrifuge at 12,000 rpm (~13,400 × g) for 2 min; pour off the waste. Then leave the column open at room temperature for a few minutes so that any residual wash buffer in the adsorption material dries off completely. Note: the point of this step is to remove the residual wash buffer from the column — leftover ethanol will interfere with downstream enzyme reactions (restriction digestion, PCR, and the like).
10. Transfer column CB3 into a clean centrifuge tube, drip 50–200 μL of elution buffer TE onto the center of the adsorption membrane, leave it at room temperature for 2–5 min, then centrifuge at 12,000 rpm (~13,400 × g) for 2 min and collect the solution in the tube. Note: don’t elute with less than 50 μL — small volumes hurt the recovery. The pH of the eluent matters a great deal: if ddH2O is used as the eluent, keep its pH between 7.0 and 8.5, since anything below 7.0 lowers

the elution efficiency. Store the DNA product at −20°C to keep it from degrading. For a higher genomic DNA yield, the collected solution can be reloaded onto column CB3, left for 2 min at room temperature, and spun again at 12,000 rpm (~13,400 × g) for 2 min.

DNA quantification and purity check

How large the recovered genomic DNA fragments come out depends on how long the sample was stored and how much shear it went through during handling. The recovered DNA can be checked on an agarose gel, with its concentration and purity read off a UV spectrophotometer.

DNA should give a strong absorption peak at OD260; an OD260 of 1 corresponds to roughly 50 μg/mL of double-stranded DNA, or 40 μg/mL of single-stranded DNA.

The OD260/OD280 ratio should sit between 1.7 and 1.9. If the DNA was eluted in ddH2O rather than elution buffer, the ratio tends to run low, because pH and ions shift the absorbance values — that doesn't by itself mean the DNA is impure.

Quantitative Real-Time PCR (qPCR)

To measure the copy number of the target gene in the D. salina genome, we ran qPCR.

This section includes:

1. Genomic DNA elimination reaction
2. Reverse transcription reaction
3. Subsequent qPCR

Protocol –Quantitative Real-Time PCR (qPCR)

cDNA was synthesized from total RNA using PrimeScript™ FAST RT reagent Kit with gDNA Eraser (RR092A, Takara) following manufacturer's instructions.

Genomic DNA Elimination Reaction

Component	Volume
8× gDNA Eraser Premix	2 μL
Genomic RNA template	1 μL
RNase Free H_2O	13 μL
Total volume	16 μL

Set up the reactions on ice and out of the light — pure water and primers go in first, with the PCR thermal profile: 42 ℃ for 2 min (alternatively 5 min at room temperature; 42 ℃ is recommended). Place on ice immediately after reaction.

Reverse Transcription Reaction

Add 4 μL of 5× RT Premix directly into the 16 μL gDNA-eliminated product from the previous step. Gently mix by pipetting and spin down briefly.

Reagent	Volume
gDNA elimination product (16 μL from last step)	16 μL
5× RT Premix	4 μL
Total volume	20 μL

RT thermal profile:

- 37 ℃, 10 min (fast reverse transcription, characteristic of this kit)
- 85 ℃, 5 sec (heat inactivation of RT enzyme to terminate reaction)
- Hold at 4 ℃

The product is cDNA. It can be used directly for qPCR or diluted with EASY Dilution II before loading. Store short-term at −20 ℃, aliquot and store long-term at −80 ℃; avoid repeated freeze-thaw cycles.

Subsequent qPCR

Prepare the reaction mix on ice. Set 3 technical replicates per sample and NTC (no-template control).

Reagent	Volume
TB Green Premix Ex Taq II (2×)	10 μL
Forward Primer (10 μM)	0.8 μL
Reverse Primer (10 μM)	0.8 μL
cDNA template	2μL
RNase Free H_2O	6.4 μL

Total volume	20 μL

qPCR amplification program (compatible with QuantStudio and other ABI instruments)

1. Initial denaturation: 95 ℃, 30 s
2. Amplification (40 cycles): 95 ℃ for 5 s; 60 ℃ for 30 s (combined annealing & extension; adjust temperature according to primer Tm)
3. Melt curve analysis: 95 ℃ for 15 s → 60 ℃ for 1 min → ramp slowly to 95 ℃ with fluorescence acquisition to verify amplification specificity

Data processing

1. Export the raw Ct values, melt curves, and amplification curves once the run is over.
2. Look over the melt curves: a single, sharp, symmetric peak means specific amplification; data with extra peaks or primer dimers don't count as valid.
3. Pool the Ct values from each group, work out the means and standard deviations, and trim outliers.
4. Use an endogenous D. salina housekeeping gene as the internal reference to correct for DNA loading and extraction variation across samples.
5. Convert to relative genome copy numbers of the exogenous target gene with the $2^{-\Delta\Delta Ct}$ method.
6. Compare the signal between wild-type and transgenic lines to rank them by exogenous gene copy number, and pick out the stable, high-expression engineered strains.

Thin-Layer Chromatography for Product Verification

In this chapter, normal-phase silica thin-layer chromatography (TLC) was used for rapid screening of crocins/astaxanthin cleavage products in engineered algae. Target analytes included crocetin, astaxanthin and related aldehyde derivatives. TLC served as a semi-quantitative screening method. Positive results were further validated by HPLC-DAD or LC-MS.

This section includes:

1. Pigment Extraction and Sample Preparation
2. TLC Separation Procedure
3. Visualization and Qualitative Identification

Protocol –Thin-Layer Chromatography for Product Verification

Pigment Extraction and Sample Preparation

1. Collect algal pellets from 1–5 mL algal culture by centrifugation at 4000×g for 5 min at 4 °C, and wash the pellets once with 0.5 M NaCl solution.
2. Add 0.5 mL of 90% aqueous acetone containing 0.01% BHT, and perform ice-bath sonication for 10 min for extraction.
3. Centrifuge at 12000 rpm for 5 min and collect the supernatant. Re-extract the pellet once, then combine all supernatants.
4. Add an equal volume of n-hexane (pre-washed with saturated NaCl solution) to extract carotenoids, and collect the upper organic phase.
5. Dry the organic phase over anhydrous Na_2SO_4, concentrate under nitrogen flow, reconstitute the residue in 50–100 μL acetone, and store at −20 °C in the dark.

TLC Separation Procedure

1. TLC plate: Silica gel GF254 aluminum-backed plate (10 × 10 cm). Activate the plate at 105 °C for 30 min before use, then equilibrate in a desiccator.
2. Spotting: Lightly draw a pencil line 1.2 cm above the bottom edge. Spot samples with capillary tubes; the spot diameter should be ⩽ 3 mm. Load 0.1 – 0.5 μ g per spot for standards and 5 – 15 μ L for samples.
3. Mobile phase:

   Crocetin: ethyl acetate : methanol : formic acid = 75:25:1

   Astaxanthin: petroleum ether (60–90 °C fraction) : acetone = 6:4 Pre-equilibrate the developing chamber for 20–30 min.
4. Development: Remove the plate when the solvent front reaches approximately 1 cm from the top (~8 cm migration distance). Immediately mark the solvent front with a pencil and air-dry the plate in the dark.

Visualization and Qualitative Identification

1. Observe the color and position of crocetin and astaxanthin bands under visible light.
2. Examine quenched fluorescent bands of GF254 under 254 nm UV light.